Daily Number Theory Drill #3

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Cyclic Patterns · Telescoping Exponents · Last Digit Sequences.

1. Find the remainder when $10^{10} + 10^5 + 1$ is divided by 9.

Ans: 3

2. What is the unit digit of 3^{2026}?

Ans: 9

3. Find the largest prime factor of 1001.

Ans: 13

4. Find the remainder when $7 \times 8 \times 9$ is divided by 5.

Ans: 4

5. If n is an integer such that $2n \equiv 3 \pmod 5$, what is the smallest positive value of n?

Ans: 4

6. Evaluate $\gcd(144, 84)$.

Ans: 12

7. Find the number of positive divisors of 36.

Ans: 9

8. Find the remainder when 2^{10} is divided by 11.

Ans: 1

9. Express 120 as a product of prime factors.

Ans: $2^3 \times 3 \times 5$

10. Find the smallest positive integer k such that $12k$ is a perfect square.

Ans: 3

11. How many of the numbers $3, 4, 5, 6, 7$ are factors of the sum $3^{2026} + 3^{2025} + 3^{2024}$?

Ans: 1

12. Find the highest power of 2 that divides 100!.

Ans: 97

13. If a and b are non-zero digits, find the maximum possible integer value of $\dfrac{10a+b}{a+b}$.

Ans: 9

14. Find the number of integer pairs (x, y) such that $xy = 24$.

Ans: 16

15. What is the smallest positive integer greater than 2 that leaves a remainder of 2 when divided by $3, 4$, and 5?

Ans: 62

16. Find the remainder when $11^{11} + 12^{12}$ is divided by 10.

Ans: 7

17. Find the value of p if p and $p^2 + 2$ are both prime numbers.

Ans: 3

18. Given that n is an integer, for how many values of n is $\dfrac{2n + 15}{n + 2}$ an integer?

Ans: 4

19. Find the sum of all positive integers n for which $n^2 - 1$ is a prime number.

Ans: 2

20. Determine the two last digits of $5^{2026} + 6^{2026}$.

Ans: 81

21. Find all pairs of positive integers (x, y) such that $\dfrac{1}{x} + \dfrac{1}{y} = \dfrac{1}{3}$.

Ans: $(4, 12), (6, 6), (12, 4)$

22. Solve the equation $x^2 - y^2 = 2025$ for positive integers (x, y) such that y is as small as possible.

Ans: $x = 51, y = 24$

23. Find all pairs of integers (x, y) that satisfy $xy + x + y = 14$.

Ans: $(0, 14), (14, 0), (-2, -16), (-16, -2), (2, 4), (4, 2), (-4, -6), (-6, -4)$

24. Find all integers x such that $x^4 + x^2 + 1$ is a prime number.

Ans: $1, -1$

25. How many ways can we choose the digits a and b such that the 5-digit number $3a4b2$ is divisible by 36?

Ans: 6

26. Find the number of 3-element subsets of $\{1, 2, \ldots, 10\}$ such that the sum of the elements is divisible by 3.

Ans: 42

27. Find the sum of all integer values of x for which $\dfrac{x^2 + 7x + 2}{x + 1}$ is an integer.

Ans: −6

28. Find the last two digits of 2019^{2025}.

Ans: 99

29. Find the number of pairs of positive integers (a, b) with $a \leq b$ such that $\text{lcm}(a, b) = 2^3 \cdot 3^2 \cdot 5$.

Ans: 53

30. Find all pairs of prime numbers (p, q) such that $p^2 - 2q^2 = 1$.

Ans: (3, 2)

31. Find the number of integer solutions to $x^3 + 2y^3 = 4z^3$.

Ans: 1

32. Find the last three digits of 7^{9999}.

Ans: 143

33. Find the last three digits of 2025^{2026}.

Ans: 625